

IEOR E4004

Optimization Models and Methods

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Outline

- Introduction to Duality Theory
- Duality Theorems
 - Weak Duality
 - Strong Duality
 - Complementary Slackness
- Economic Interpretation of the Dual Problem

Introduction to Duality Theory

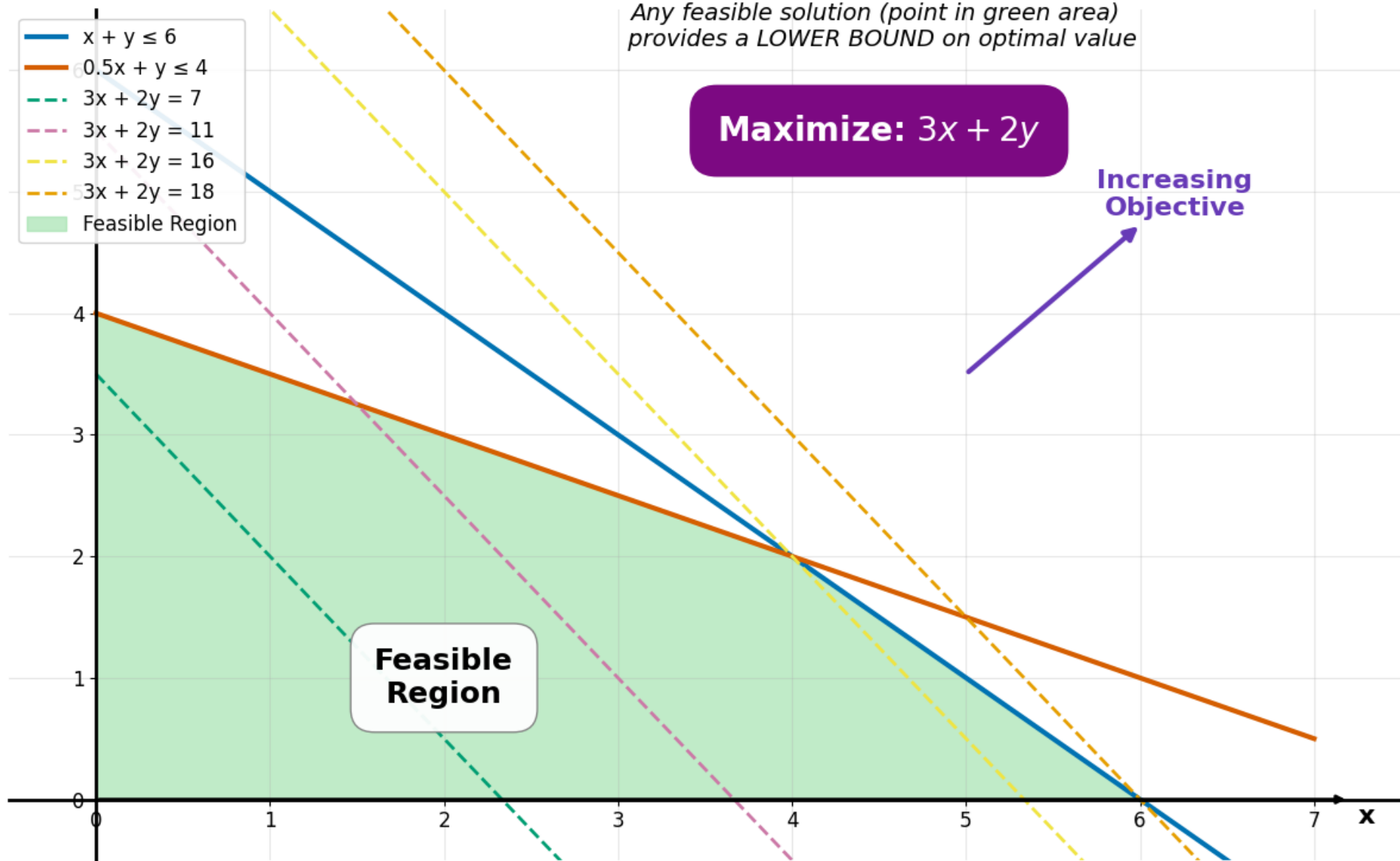
Bound On Optimal Value

When dealing with a maximizing linear program, any feasible solution gives a lower, or suboptimal, bound on the optimal value.

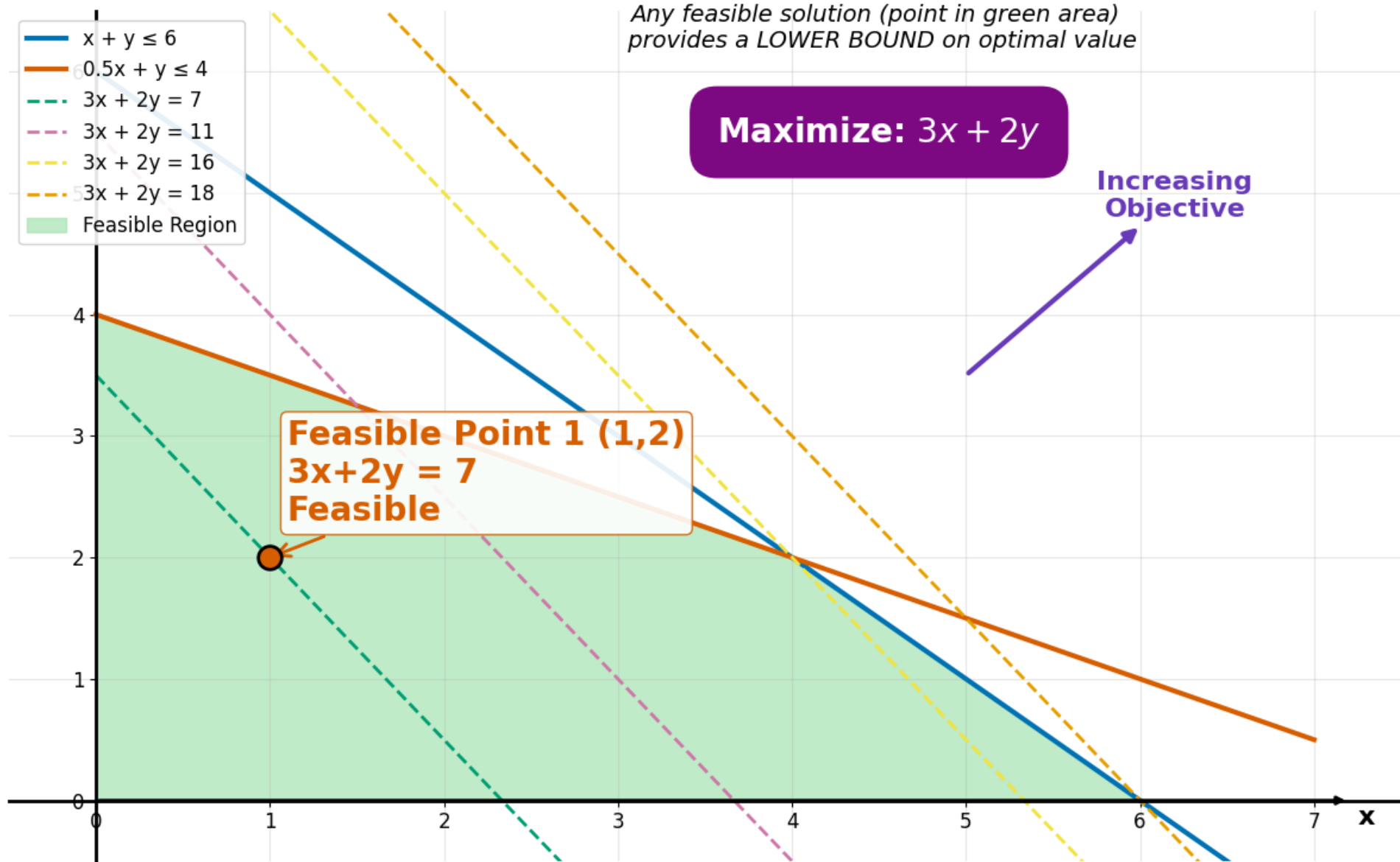
Let's first consider this simple 2 variable LP as an example:

$$\begin{array}{ll}\text{Maximize} & 3x + 2y \\ \text{subject to} & x + y \leq 6 \\ & 0.5x + y \leq 4 \\ & x \geq 0 \\ & y \geq 0\end{array}$$

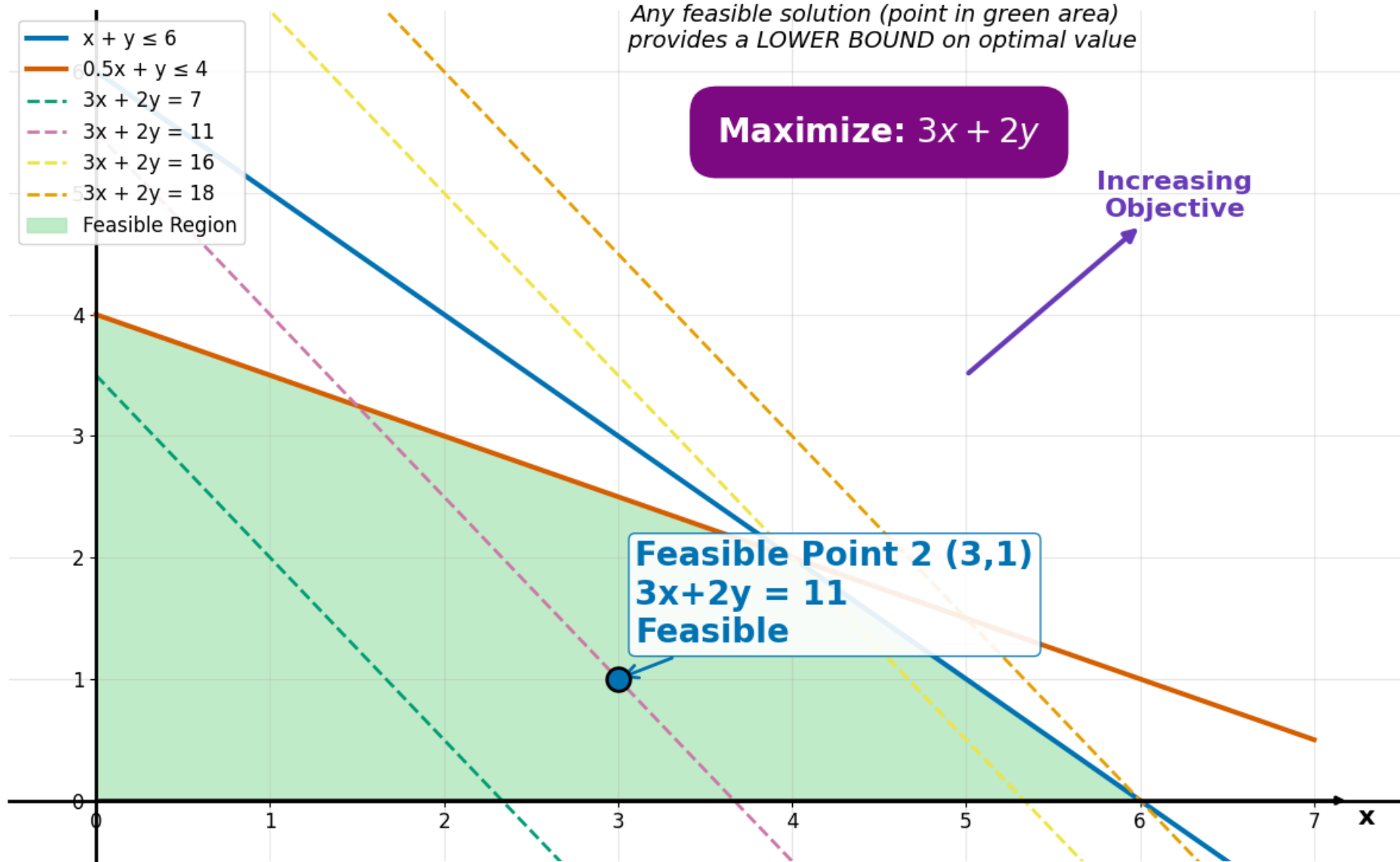
y Feasible Solutions as Lower Bounds (Maximization LP)



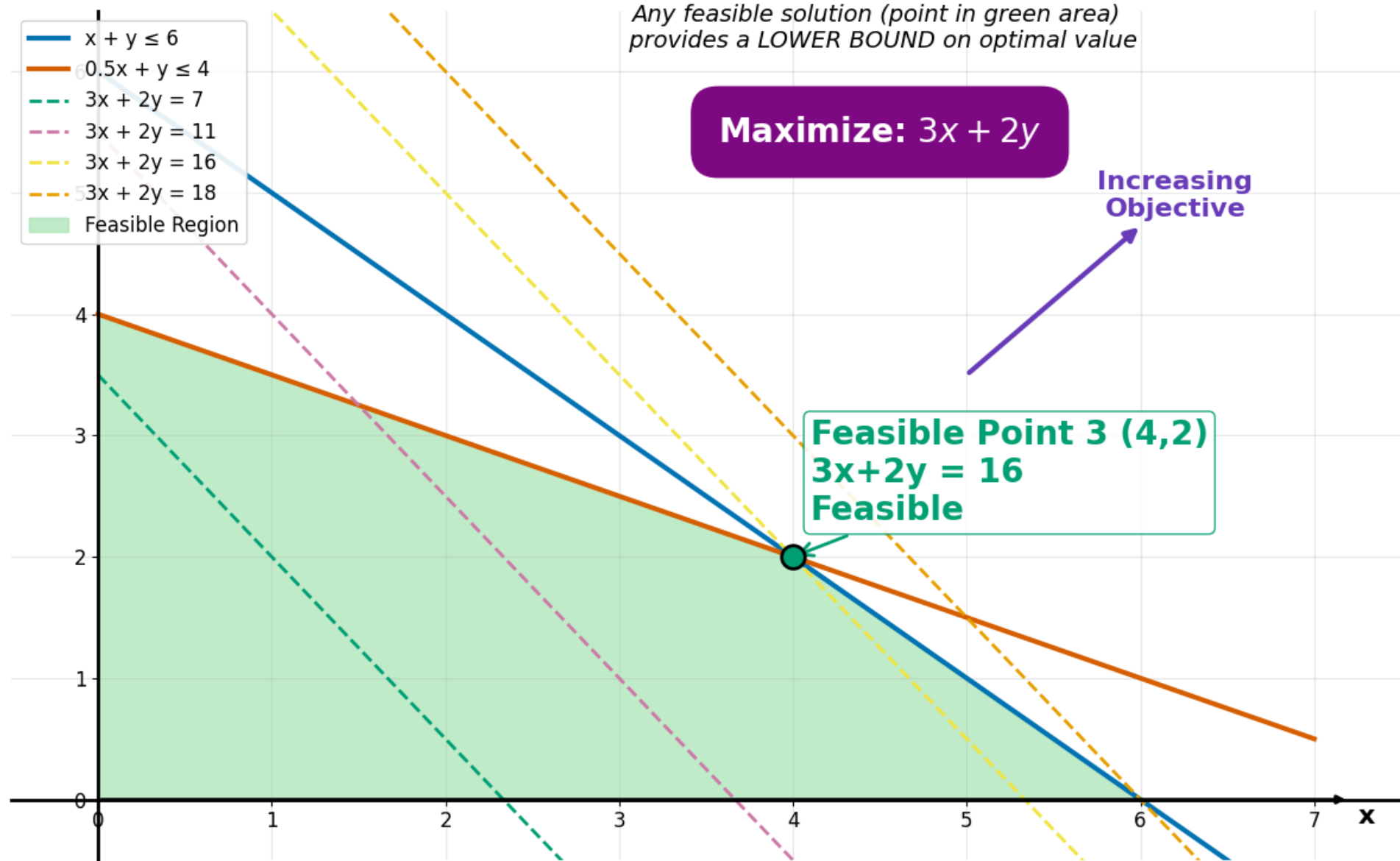
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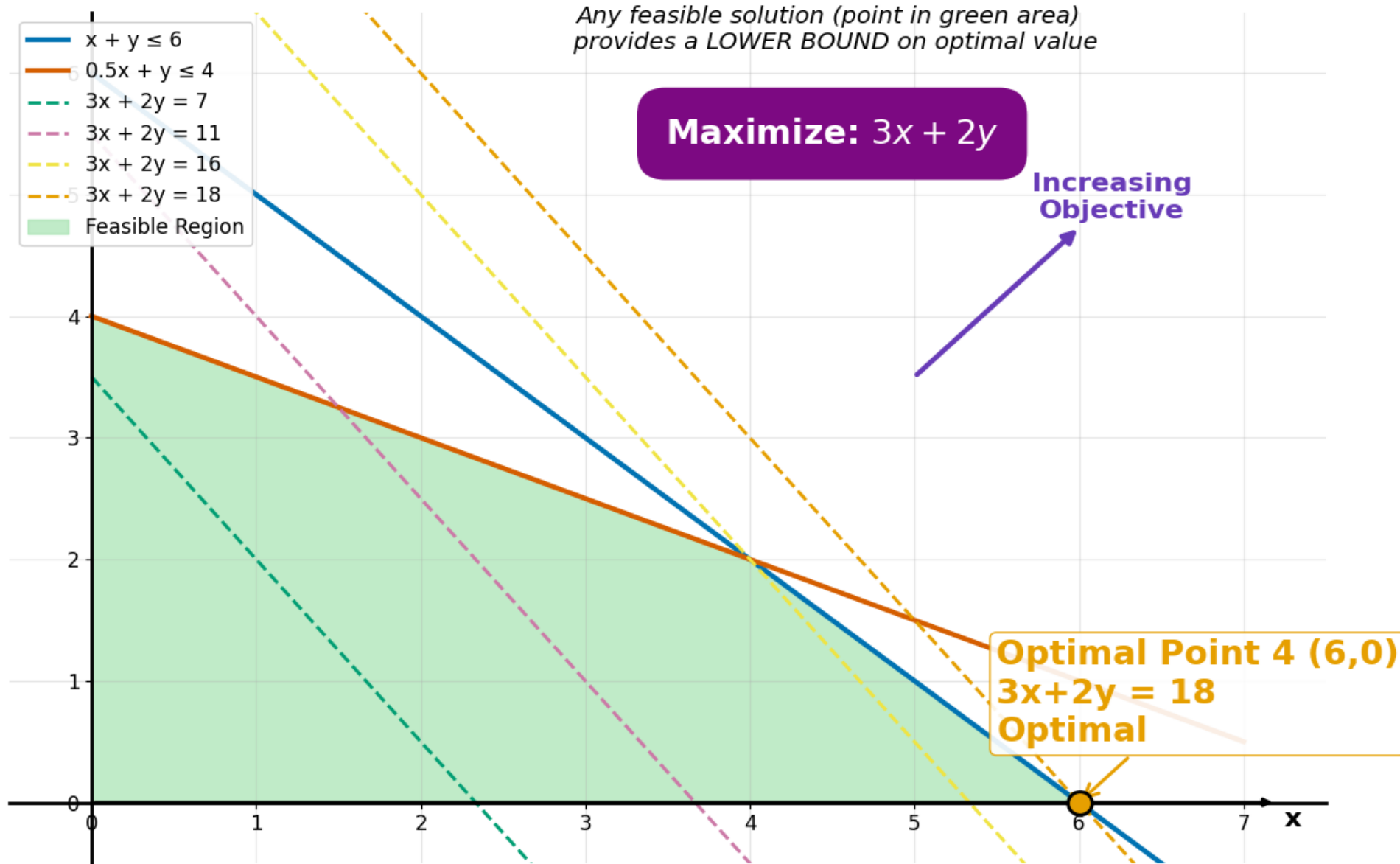
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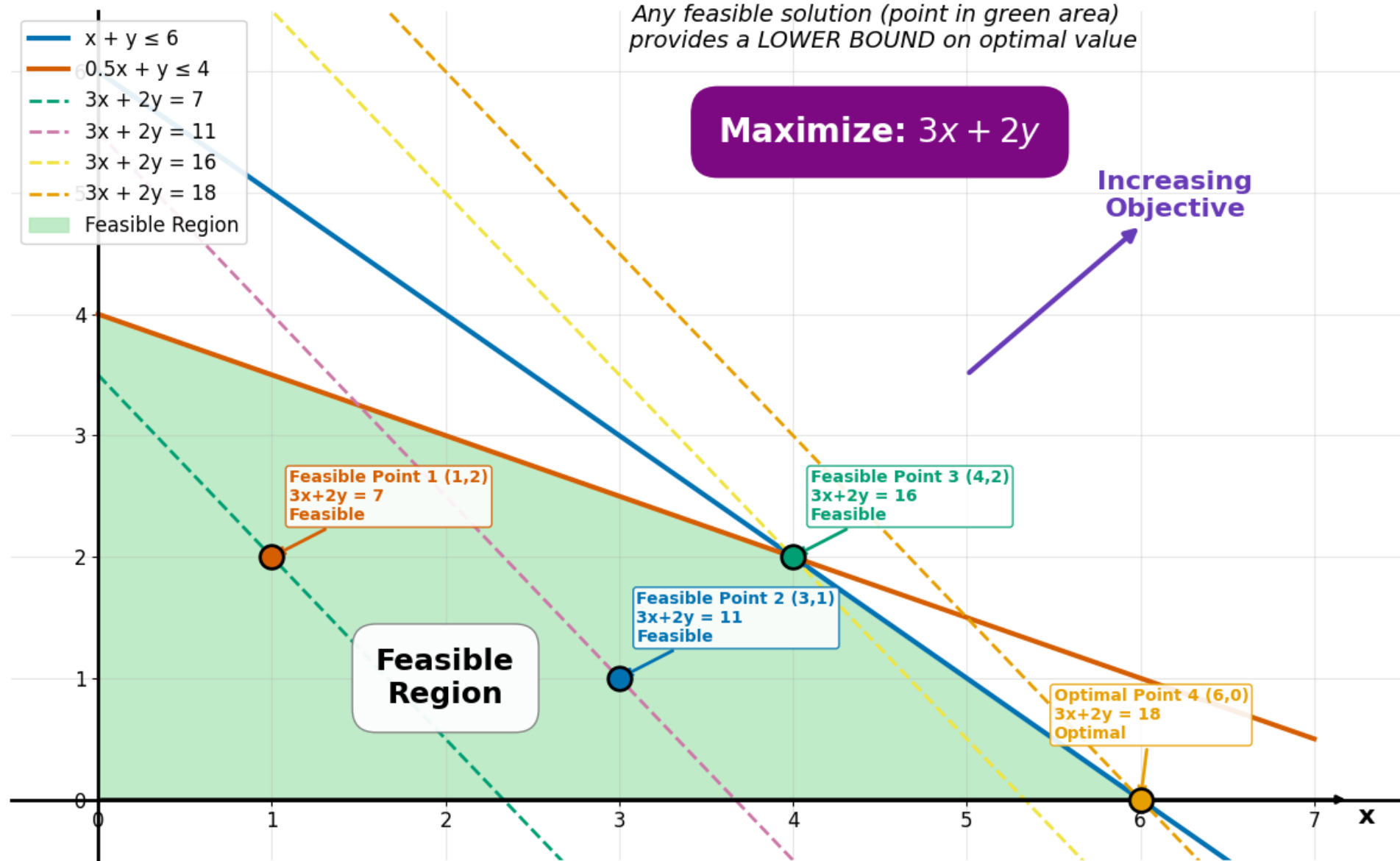
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y Feasible Solutions as Lower Bounds (Maximization LP)



Question

When dealing with a maximizing linear program, any feasible solution gives a lower, or suboptimal, bound on the optimal value.

What about finding the upper bounds?

Is This Linear Program (LP) Bounded?

$$\begin{array}{ll}\max_{x_1, x_2 \in \mathbb{R}} & 3.5x_1 + 2x_2 \\ \text{s.t.} & \left\{ \begin{array}{l} x_1 + x_2 \leq 80 \\ 2x_1 + x_2 \leq 100 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{array} \right.\end{array}$$

Is This Linear Program (LP) Bounded?

$$\begin{array}{ll}\max_{x_1, x_2 \in \mathbb{R}} & 3.5x_1 + 2x_2 \\ \text{s.t.} & \begin{cases} x_1 + x_2 \leq 80 \\ 2x_1 + x_2 \leq 100 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{cases}\end{array}$$

Yes! Can I find an upper bound?

Is This Linear Program (LP) Bounded?

$$\begin{array}{ll}\max_{x_1, x_2 \in \mathbb{R}} & 3.5x_1 + 2x_2 \\ \text{s.t.} & \begin{cases} x_1 + x_2 \leq 80 \\ 2x_1 + x_2 \leq 100 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{cases} \end{array}$$

× 2
× 1

$$\begin{array}{rcl} & 2x_1 + 2x_2 \leq 160 & \\ + & 2x_1 + x_2 \leq 100 & \\ \hline 3.5x_1 + 2x_2 & \leq & 4x_1 + 3x_2 \leq 260 \end{array}$$

Is This Linear Program (LP) Bounded?

$$\begin{array}{ll}\max_{x_1, x_2 \in \mathbb{R}} & 3.5x_1 + 2x_2 \\ \text{s.t.} & \begin{cases} x_1 + x_2 \leq 80 \\ 2x_1 + x_2 \leq 100 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{cases} \end{array}$$

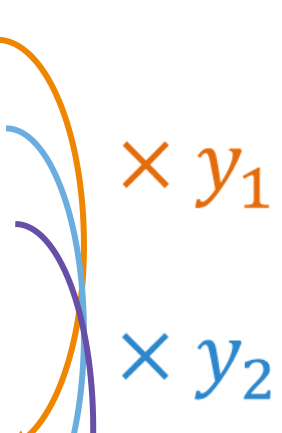
$\times 2$
 $\times 1$

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An upper bound

Can I Get A Better Bound?

$$\begin{array}{ll} \max_{x_1, x_2 \in \mathbb{R}} & 3.5x_1 + 2x_2 \\ \text{s.t.} & \begin{cases} x_1 + x_2 \leq 80 \\ 2x_1 + x_2 \leq 100 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{cases} \end{array}$$



$$\begin{array}{l} y_1 x_1 + y_1 x_2 \leq 80 y_1 \\ 2y_2 x_1 + y_2 x_2 \leq 100 y_2 \\ + \quad y_3 x_1 \leq 40 y_3 \end{array}$$

Now, we created a new problem!

$$3.5x_1 + 2x_2 \leq (y_1 + 2y_2 + y_3)x_1 + (y_1 + y_2)x_2 \leq 80y_1 + 100y_2 + 40y_3$$

Dual Linear Program

Dual Of The LP

$$3.5x_1 + 2x_2 \leq (y_1 + 2y_2 + y_3)x_1 + (y_1 + y_2)x_2 \leq 80y_1 + 100y_2 + 40y_3$$

$$\begin{array}{ll} \min_{y_1, y_2, y_3 \in \mathbb{R}} & 80y_1 + 100y_2 + 40y_3 \\ \text{s.t.} & \begin{cases} y_1 + 2y_2 + y_3 \geq 3.5 \\ y_1 + y_2 \geq 2 \\ y_1, y_2, y_3 \geq 0 \end{cases} \end{array}$$

This new LP we produced is called the **DUAL** of the LP, which makes the original problem the **PRIMAL**

Dual Of The LP

$$3.5x_1 + 2x_2 \leq (y_1 + 2y_2 + y_3)x_1 + (y_1 + y_2)x_2 \leq 80y_1 + 100y_2 + 40y_3$$

$$\begin{array}{ll} \min_{y_1, y_2, y_3 \in \mathbb{R}} & 80y_1 + 100y_2 + 40y_3 \\ \text{s.t.} & \begin{cases} y_1 + 2y_2 + y_3 \geq 3.5 \\ y_1 + y_2 \geq 2 \\ y_1, y_2, y_3 \geq 0 \end{cases} \end{array}$$

Upper bound on
the primal objective



This new LP we produced is called the **DUAL** of the LP,
which makes the original problem the **PRIMAL**

Primal And Dual LPs In Standard Matrix Form

Primal:

$$\begin{array}{ll} \max_{x_1, x_2 \in \mathbb{R}} & 3.5x_1 + 2x_2 \\ \text{s.t.} & \begin{cases} x_1 + x_2 \leq 80 \\ 2x_1 + x_2 \leq 100 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{cases} \end{array}$$

Dual:

$$\begin{array}{ll} \min_{y_1, y_2, y_3 \in \mathbb{R}} & 80y_1 + 100y_2 + 40y_3 \\ \text{s.t.} & \begin{cases} y_1 + 2y_2 + y_3 \geq 3.5 \\ y_1 + y_2 \geq 2 \\ y_1, y_2, y_3 \geq 0 \end{cases} \end{array}$$

Primal and Dual LPs in Standard Matrix Form

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$$\begin{array}{ll} \min_{y_1, y_2, y_3 \in \mathbb{R}} & 80y_1 + 100y_2 + 40y_3 \\ \text{s.t.} & \begin{cases} y_1 + 2y_2 + y_3 \geq 3.5 \\ y_1 + y_2 \geq 2 \\ y_1, y_2, y_3 \geq 0 \end{cases} \end{array}$$

Standard
Matrix
Form:

$$(P) : \max_{x \in \mathbb{R}^n} c^T x \quad \text{s.t.} \quad Ax \leq b \quad \text{and} \quad x \geq 0$$

$$(D) : \min_{y \in \mathbb{R}^m} b^T y \quad \text{s.t.} \quad y \geq 0 \quad \text{and} \quad A^T y \geq c$$

How To Find The Dual Of An LP

In general, for maximization problem with $\leq$ inequalities, the dual is obtained simply by:

- transposing (flipping around the diagonal) the matrix A
- swapping vectors b and c
- switching the inequalities to $\geq$
- changing max to min

$$\begin{array}{ll} \text{(P)} & \\ \max & \mathbf{c}^T \mathbf{x} \\ & \mathbf{Ax} \leq \mathbf{b} \\ & \mathbf{x} \geq 0 \end{array}$$

$$\begin{array}{ll} \text{(D)} & \\ \min & \mathbf{b}^T \mathbf{y} \\ & \mathbf{A}^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq 0 \end{array}$$

Example | How To Find The Dual Of An Lp

- In summary:

Primal (Max)	Dual (Min)
i -th constraint $\leq$	variable $y_i \geq 0$
i -th constraint $\geq$	variable $y_i \leq 0$
i -th constraint $=$	variable y_i unrestricted
$x_i \geq 0$	i -th constraint $\geq$
$x_i \leq 0$	i -th constraint $\leq$
x_i unrestricted	i -th constraint $=$

Example | How To Find The Dual Of An Lp

- In summary:

Primal (Max)	Dual (Min)
i -th constraint $\leq$	variable $y_i \geq 0$
i -th constraint $\geq$	variable $y_i \leq 0$
i -th constraint $=$	variable y_i unrestricted
$x_i \geq 0$	i -th constraint $\geq$
$x_i \leq 0$	i -th constraint $\leq$
x_i unrestricted	i -th constraint $=$

$$\begin{array}{rcll}
 \text{Max} & 3x_1 & + & 2x_2 & + & x_3 & & \\
 & x_1 & + & x_2 & + & \frac{1}{2}x_3 & \leq & 80 \\
 & 2x_1 & + & x_2 & + & x_3 & = & 100 \\
 & x_1 & & & + & x_3 & \geq & 40 \\
 & & & & & x_1 & \text{URS} & \\
 & & & & & x_2 & \leq & 0 \\
 & & & & & x_3 & \geq & 0
 \end{array}$$

? Can you find the dual of this LP?

Example | How To Find The Dual Of An Lp

$$\begin{array}{llllll} \text{Max} & 3x_1 & + & 2x_2 & + & x_3 \\ & x_1 & + & x_2 & + & \frac{1}{2}x_3 & \leq & 80 \\ & 2x_1 & + & x_2 & + & x_3 & = & 100 \\ & x_1 & & & & + & x_3 & \geq & 40 \\ & & & & & x_1 & \text{URS} \\ & & & & & x_2 & \leq & 0 \\ & & & & & x_3 & \geq & 0 \end{array}$$

Primal

$$\begin{array}{llllll} \text{Min} & 80y_1 & + & 100y_2 & + & 40y_3 \\ & y_1 & + & 2y_2 & + & y_3 & = & 3 \\ & y_1 & + & y_2 & & & \leq & 2 \\ & \frac{1}{2}y_1 & + & y_2 & + & y_3 & \geq & 1 \\ & & & & & y_1 & \geq & 0 \\ & & & & & y_2 & \text{URS} \\ & & & & & y_3 & \leq & 0 \end{array}$$

Dual

Duality Theorems

Duality Theorems – Weak Duality

1. Weak Duality Theorem

! If x' is any **feasible solution** of a maximization primal and y' is any **feasible solution** of the dual, then $val(P) \leq val(D)$,
(can also be written as: $c^T x' \leq b^T y'$)

- In other words, the value of any feasible solution to the **dual** yields an **upper bound** on the value of any feasible solution (including the optimal) to the **primal**

Example | 2-Variable Example

$$\begin{array}{ll}\text{Maximize} & z = 3x_1 + 2x_2 \\ \text{subject to} & x_1 + x_2 \leq 6 \\ & 0.5x_1 + x_2 \leq 4 \\ & x_1 \geq 0, \quad x_2 \geq 0\end{array}$$

Primal

$$\begin{array}{ll}\text{Minimize} & w = 6y_1 + 4y_2 \\ \text{subject to} & y_1 + 0.5y_2 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1 \geq 0, \quad y_2 \geq 0\end{array}$$

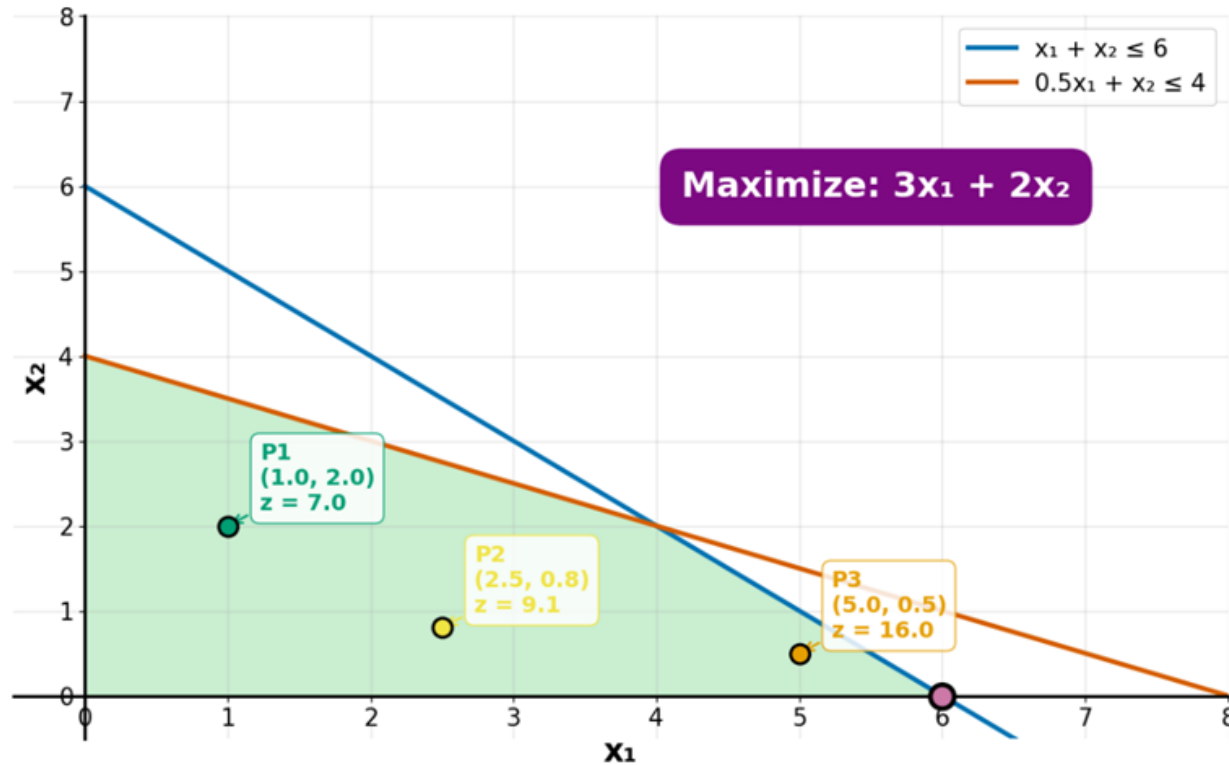
Dual

Example | Feasible Solutions Of Dual And Primal

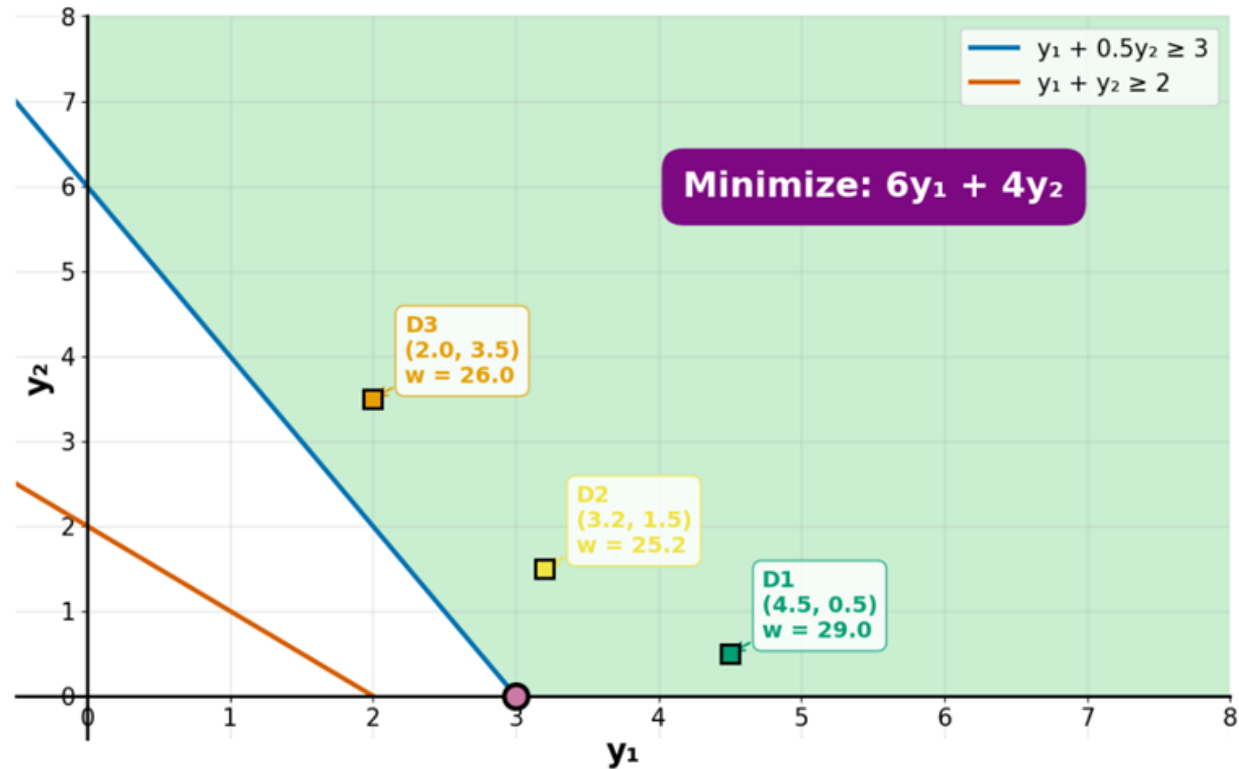
Weak Duality Theorem: Primal $\leq$ Dual for All Feasible Solutions

Any primal feasible solution provides a LOWER BOUND | Any dual feasible solution provides an UPPER BOUND

PRIMAL: Maximize $3x_1 + 2x_2$



DUAL: Minimize $6y_1 + 4y_2$



Duality Theorems – Weak Duality

1. Hence, if **primal is unbounded**, then **dual must be infeasible**
2. Likewise, if **dual is unbounded**, then **primal must be infeasible**
3. Note that it is possible that both primal and dual are infeasible
4. But if both are feasible, then neither of them is unbounded

		primal		
		infeasible	feasible bounded	unbounded
dual	infeasible	✓ (3)	✗	✓ (1)
	feasible bounded	✗	✓ (4)	✗
	unbounded	✓ (2)	✗	✗

Duality Theorems – Strong Duality

1. Strong Duality Theorem

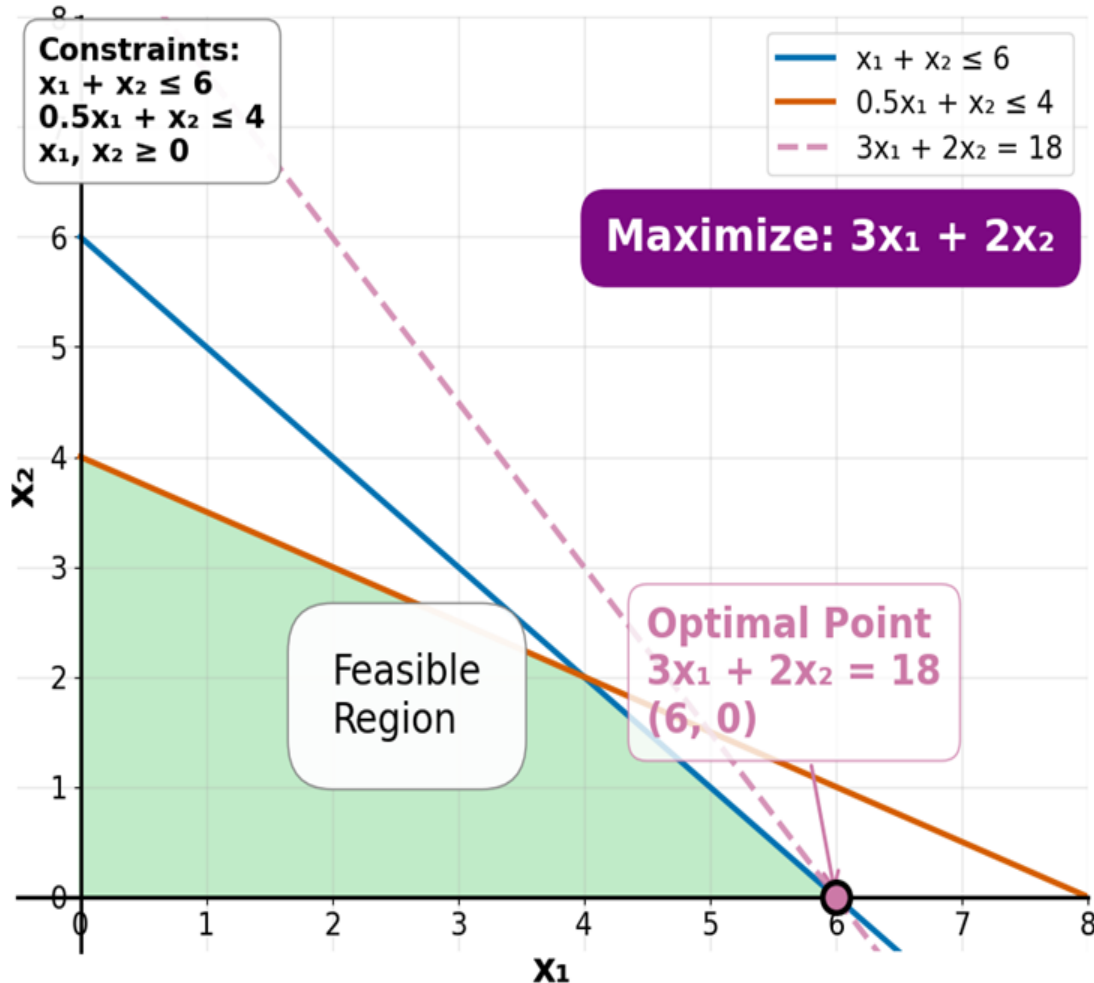
! If x^* is an optimal solution to the primal and y^* is an optimal solution to the dual, then $val(P) = val(D)$
(can also be written as: $c^T x^* = b^T y^*$)

- In other words, if the **dual** yields to an optimal solution, the **primal** also will have the same optimal value.

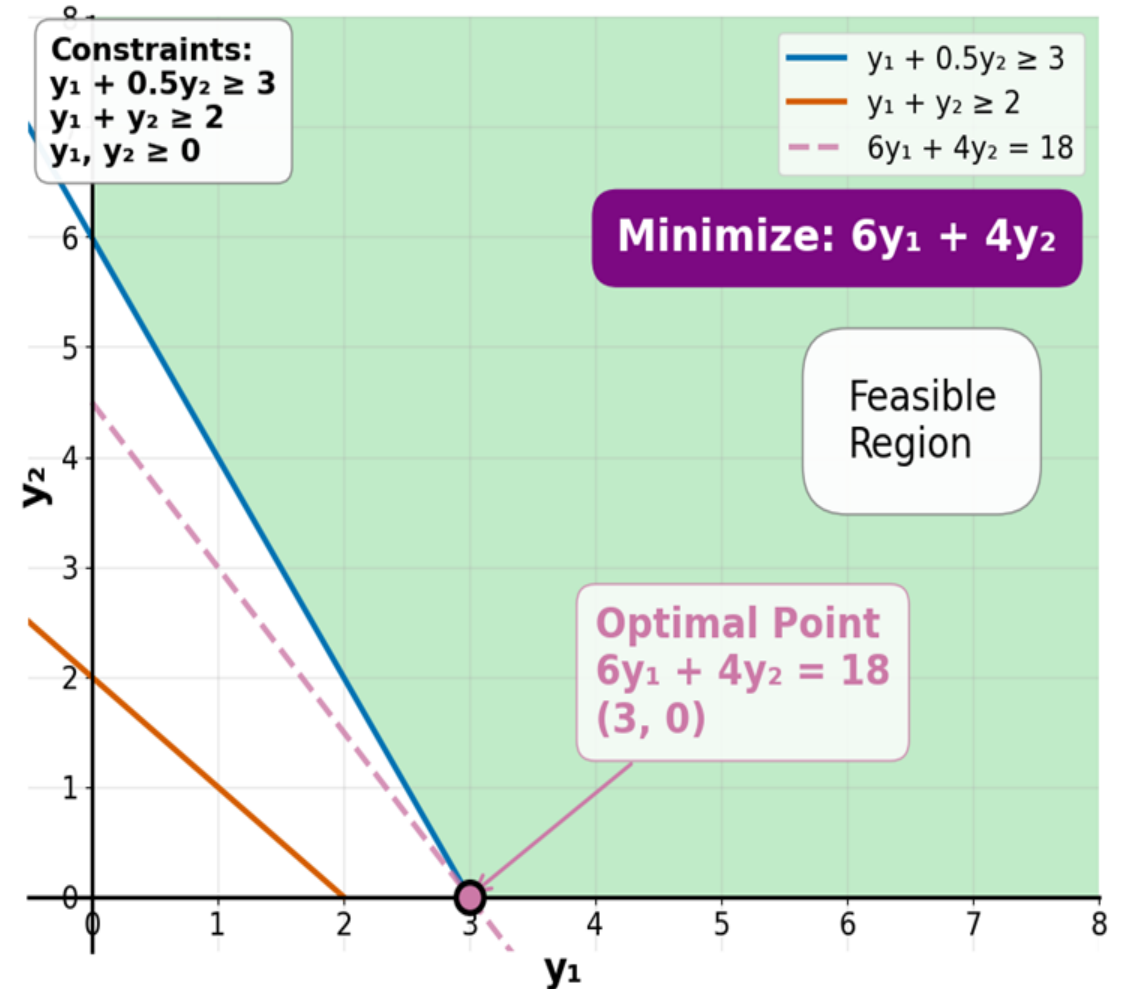
Example | Optimal Solutions Of Dual And Primal

Strong Duality Theorem: Optimal Values are Equal

Primal Problem: Maximize $3x_1 + 2x_2$



Dual Problem: Minimize $6y_1 + 4y_2$



Duality Theorems – Complementary Slackness

3. Complementary Slackness Theorem

- At **optimality**, if a constraint is not tight (has a positive slack) in the primal, then the corresponding dual variable must be 0:

$$A_i x \neq b_i \rightarrow y_i = 0$$

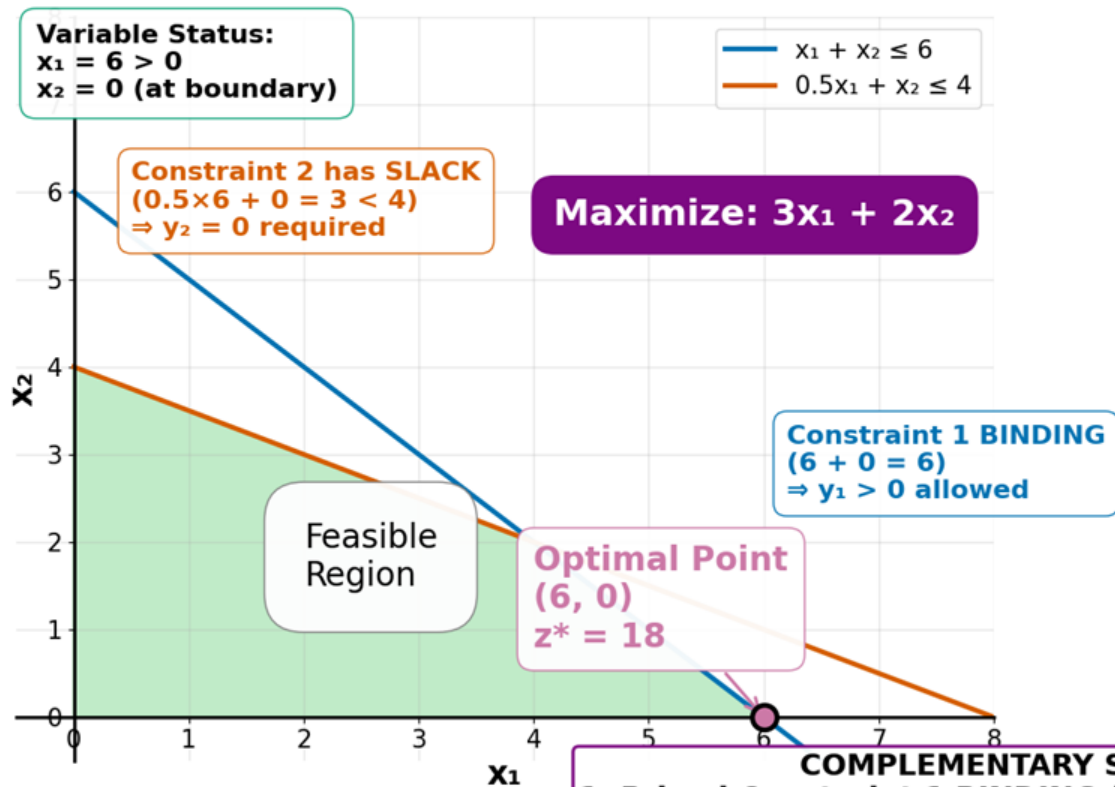
- Conversely, if a primal variable is strictly positive, then the corresponding dual constraint must be tight (i.e., slack is 0):

$$x_i > 0 \rightarrow A_i^T y = c_i$$

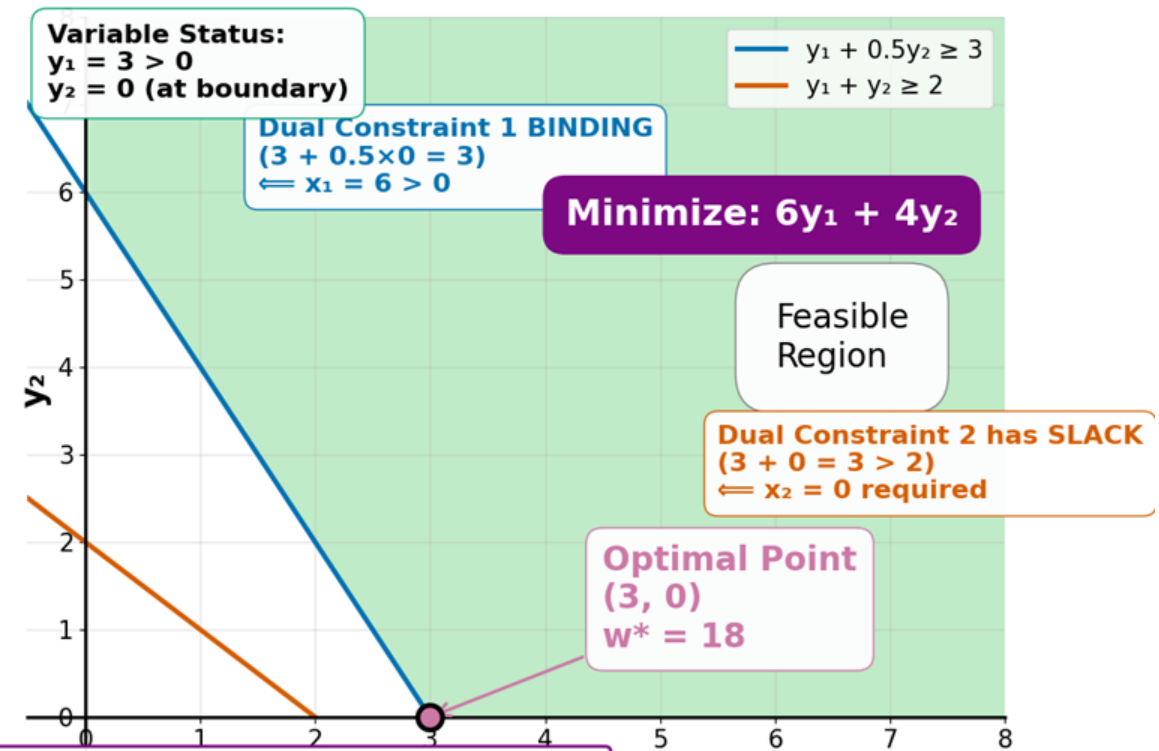
Example | Complementary Slackness

Complementary Slackness Conditions in Linear Programming
Strong Duality: $z^* = w^* = 18$ | Both problems achieve the same optimal value

PRIMAL: Maximize $3x_1 + 2x_2$



DUAL: Minimize $6y_1 + 4y_2$



COMPLEMENTARY SLACKNESS CONDITIONS:

1. Primal Constraint 1 BINDING (slack = 0) $\Leftrightarrow$ Dual Variable $y_1 = 3 > 0$
2. Primal Constraint 2 has SLACK (= 1) $\Leftrightarrow$ Dual Variable $y_2 = 0$
3. Dual Constraint 1 BINDING (slack = 0) $\Leftrightarrow$ Primal Variable $x_1 = 6 > 0$
4. Dual Constraint 2 has SLACK (= 1) $\Leftrightarrow$ Primal Variable $x_2 = 0$

Duality Theorems

Using Complementary Slackness To Evaluate Optimality

- Consider this LP:

Primal

$$\begin{aligned} \max z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad &2x_1 + x_2 \leq 100 \\ &x_1 + x_2 \leq 80 \\ &x_1 \leq 40 \\ &x_1, x_2 \geq 0 \end{aligned}$$

Dual

$$\begin{aligned} \min W &= 100y_1 + 80y_2 + 40y_3 \\ \text{s.t.} \quad &2y_1 + y_2 + y_3 \geq 3 \\ &y_1 + y_2 \geq 2 \\ &y_1, y_2, y_3 \geq 0 \end{aligned}$$

Using Complementary Slackness To Evaluate Optimality



Without solving the primal or dual, can you tell whether $x_1 = 10$ and $x_2 = 30$ is optimal?

Using Complementary Slackness To Evaluate Optimality

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Yes! Using complementary slackness, we can evaluate optimality.

Using Complementary Slackness To Evaluate Optimality



Without solving the primal or dual, can you tell whether $x_1 = 10$ and $x_2 = 30$ is optimal?

Yes! Using complementary slackness, we can evaluate optimality.

Remember, at optimality:

- For each non-binding (positive slack) constraint in the primal, the corresponding dual variable is zero.
- For each primal variable that is zero, the corresponding dual constraint is binding.

Using Complementary Slackness To Evaluate Optimality

Primal

$$\begin{aligned} \max z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad 2x_1 + x_2 &\leq 100 \\ x_1 + x_2 &\leq 80 \\ x_1 &\leq 40 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Check if $x_1^* = 10$ and $x_2^* = 30$ is optimal

Using Complementary Slackness To Evaluate Optimality

Primal

$$\begin{array}{ll}\max z = & 3x_1 + 2x_2 \\ \text{s.t.} & 2x_1 + x_2 \leq 100 \\ & x_1 + x_2 \leq 80 \\ & x_1 \leq 40 \\ & x_1, x_2 \geq 0\end{array}$$

Check if $x_1^* = 10$ and $x_2^* = 30$ is optimal

In the Primal LP:

1. $2(10) + 30 = 50 < 100$ (*not binding*)
2. $10 + 30 = 40 < 80$ (*not binding*)
3. $10 < 40$ (*not binding*)

Using Complementary Slackness To Evaluate Optimality

Dual

$$\min W = 100y_1 + 80y_2 + 40y_3$$

$$\text{s.t.} \quad 2y_1 + y_2 + y_3 \geq 3$$

$$y_1 + y_2 \geq 2$$

$$y_1, y_2, y_3 \geq 0$$

According to complementary slackness, all corresponding dual variables $y_1, y_2, y_3 = 0$.

Using Complementary Slackness To Evaluate Optimality

Dual

$$\begin{array}{ll}\min W = & 100y_1 + 80y_2 + 40y_3 \\ \text{s.t.} & 2y_1 + y_2 + y_3 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2, y_3 \geq 0\end{array}$$

According to complementary slackness, all corresponding dual variables $y_1, y_2, y_3 = 0$.

Check the Dual LP where $y_1, y_2, y_3 = 0$:

1. $0 + 0 + 0 \geq 3$ (*not satisfied*)
2. $0 + 0 \geq 2$ (*not satisfied*)

Using Complementary Slackness To Evaluate Optimality

Dual

$$\begin{array}{ll}\min W = & 100y_1 + 80y_2 + 40y_3 \\ \text{s.t.} & 2y_1 + y_2 + y_3 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2, y_3 \geq 0\end{array}$$

According to complementary slackness, all corresponding dual variables $y_1, y_2, y_3 = 0$.

Check the Dual LP where $y_1, y_2, y_3 = 0$:

1. $0 + 0 + 0 \geq 3$ (*not satisfied*)
2. $0 + 0 \geq 2$ (*not satisfied*)

Therefore, $x_1 = 10$ and $x_2 = 30$ is not optimal!

Using Complementary Slackness To Evaluate Optimality

Primal

$$\begin{aligned} \max z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad 2x_1 + x_2 &\leq 100 \\ x_1 + x_2 &\leq 80 \\ x_1 &\leq 40 \\ x_1, x_2 &\geq 0 \end{aligned}$$

Check if $x_1^* = 20$ and $x_2^* = 60$ is optimal

Using Complementary Slackness To Evaluate Optimality

Primal

$$\begin{aligned} \max z &= 3x_1 + 2x_2 \\ \text{s.t.} \quad &2x_1 + x_2 \leq 100 \\ &x_1 + x_2 \leq 80 \\ &x_1 \leq 40 \\ &x_1, x_2 \geq 0 \end{aligned}$$

Check if $x_1^* = 20$ and $x_2^* = 60$ is optimal

In the Primal LP:

1. $2(20) + 60 = 100 = 100$ (*binding, $y_1 > 0$*)
2. $20 + 60 = 80 = 80$ (*binding, $y_2 > 0$*)
3. $20 < 40$ (*not binding, and $y_3 = 0$*)

Using Complementary Slackness To Evaluate Optimality

Dual

$$\begin{array}{ll}\min W = & 100y_1 + 80y_2 + 40y_3 \\ \text{s.t.} & 2y_1 + y_2 + y_3 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2, y_3 \geq 0\end{array}$$



Your Dual problem changes when you substitute $y_3 = 0$

$$\begin{array}{ll}\min W = & 100y_1 + 80y_2 \\ \text{s.t.} & 2y_1 + y_2 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2 > 0\end{array}$$

Using Complementary Slackness To Evaluate Optimality

Dual

$$\begin{array}{ll}\min W = 100y_1 + 80y_2 + 40y_3 \\ \text{s.t.} & 2y_1 + y_2 + y_3 \geq 3 \\ & y_1 + y_2 \geq 2 \\ & y_1, y_2, y_3 \geq 0\end{array}$$

Your Dual problem changes when you substitute $y_3 = 0$

$$\begin{array}{lll}\min W = 100y_1 + 80y_2 & \longrightarrow & \min W = 100y_1 + 80y_2 \\ \text{s.t.} & 2y_1 + y_2 \geq 3 & \longrightarrow \text{s.t.} \quad 2y_1 + y_2 = 3 \\ & y_1 + y_2 \geq 2 & y_1 + y_2 = 2 \\ & y_1, y_2 > 0 & y_1, y_2 > 0\end{array}$$

When we apply complementary slackness, the primal variables $x_1 = 20 > 0$ and $x_2 = 60 > 0$, corresponds to binding dual constraints!

Let's solve this system of linear equations

Using Complementary Slackness To Evaluate Optimality

$$\begin{array}{ll}\min W = 100y_1 + 80y_2 \\ \text{s.t.} & 2y_1 + y_2 = 3 \\ & y_1 + y_2 = 2 \\ & y_1, y_2 > 0\end{array}$$

We find $y_1 = 1$ and $y_2 = 1$, and since $y_3 = 0$, these satisfy the dual constraints, and satisfy the complementary slackness conditions:

- $x_1 = 20 > 0$, the corresponding dual constraint $2y_1 + y_2 = 3$ is binding.
- $x_2 = 60 > 0$, the corresponding dual constraint $y_1 + y_2 = 2$ is binding.
- The non-binding primal constraint corresponds to $y_3 = 0$.

Using Complementary Slackness To Evaluate Optimality

$$\begin{array}{ll}\min W = 100y_1 + 80y_2 \\ \text{s.t.} & 2y_1 + y_2 = 3 \\ & y_1 + y_2 = 2 \\ & y_1, y_2 > 0\end{array}$$

We find $y_1 = 1$ and $y_2 = 1$, and since $y_3 = 0$, these satisfy the dual constraints, and satisfy the complementary slackness conditions:

- $x_1 = 20 > 0$, the corresponding dual constraint $2y_1 + y_2 = 3$ is binding.
- $x_2 = 60 > 0$, the corresponding dual constraint $y_1 + y_2 = 2$ is binding.
- The non-binding primal constraint corresponds to $y_3 = 0$.

The solution is optimal as it satisfies the complementary slackness condition!

$$x_1^* = 20 \text{ and } x_2^* = 60$$

Economic Interpretation of the Dual Problem

Furniture Manufacturing Company



A furniture company has limited resources for production in a week.

- 100 hours of woodwork labor per week
- 80 units of assembly labor per week

Each chair requires 2 hours of labor, 4 unit of assembly labor, earning \$20 profit.

Each table requires 4 hours of woodwork labor, 2 units of assembly labor, earning \$30 profit.

Furniture Manufacturing Company



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- 80 units of assembly labor per week

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Each table requires 4 hours of woodwork labor, 2 units of assembly labor, earning \$30 profit.

Goal:

- Maximize profit by deciding how many chairs and tables to produce. (*Primal*)

Alternative Goal:

- Minimize the total cost of resources needed to support the profit you're getting from production. (*Dual*)

Furniture Manufacturing Company



Alternative goal considers the value of resources:

x_1 = number of chairs

x_2 = number of tables

y_1 = value (shadow price) of one woodwork hour

y_2 = value (shadow price) of one assembly unit

Max $Z = 30x_1 + 20x_2$

$$4x_1 + 2x_2 \leq 100 \text{ (Woodwork hours)}$$

$$2x_1 + 4x_2 \leq 80 \text{ (Assembly hours)}$$

$$x_1, x_2 \geq 0$$

Primal

Min $w = 100y_1 + 80y_2$

$$4y_1 + 2y_2 \geq 30 \text{ (value must cover profit per Table)}$$

$$2y_1 + 2y_2 \geq 20 \text{ (value must cover profit per Chair)}$$

$$y_1, y_2 \geq 0$$

Dual

Furniture Manufacturing Company



Alternative goal considers the value of resources:

y_1 = value (shadow price) of one woodwork hour

y_2 = value (shadow price) of one assembly unit

Suppose someone offered to buy all our weekly resources.
What is the *minimum total value* we should accept?

Min $w = 100y_1 + 80y_2$

$$4y_1 + 2y_2 \geq 30 \text{ (value must cover profit per Table)}$$

$$2y_1 + 2y_2 \geq 20 \text{ (value must cover profit per Chair)}$$

$$y_1, y_2 \geq 0$$

No product should generate profit larger than the cost of the resources it consumes.

Dual

Furniture Manufacturing Company



Consider $x_1 = 10$ tables and $x_2 = 15$ chairs

$$4x_1 + 2x_2 \leq 100 \text{ (Woodwork hours)}$$

$$2x_1 + 4x_2 \leq 80 \text{ (Assembly hours)}$$

From complementary slackness:

- Woodwork hours has slack = 30 and $y_1 = 0$ and
- Assembly hours are fully used, slack = 0 and $y_2 > 0$

Dual variables reflect the value of relaxing constraints. As assembly labor is fully utilized, it is a critical resource. This means that adding more hours would increase total profit.

If a resource has slack and if a product is produced, its associated dual constraint must be tight.

Announcements | Upcoming Quiz



Quiz 1 Information

Date: February 18

Time: During lecture (full lecture period)



Format

There will be **1–2 questions** focusing on:

- Formulating Linear Programs
- Solving LPs using the **graphical method**
- Discussion and evaluation of methods we have covered



What Will NOT Be Included

- No solving using the **Simplex method** (*But you should understand the logic and be able to comment on it.*)
- No programming component
- No coding (including on paper)



Rules

- Closed book & closed notes
- You may bring **one double-sided, A4 handwritten cheat sheet** (*No printed or photocopied sheets allowed*)
- Calculators are allowed



Not Allowed

- Smartphones
- Smartwatches
- Headphones